

A Reflective Journey: Navigating My Cumulative Experience at Iowa State University

I began my journey at Iowa State University in August of 2022. It was definitely a challenging experience moving away from home to start my higher education here in Ames, but one that had to be done and turned out to be a great decision. When deciding my major in high school, I knew it had to be something STEM related, and was leaning to a computer related field. I chose computer engineering because it was not heavily focused on one specific field. In my high school experience, I enjoyed learning about software development and coding, but I was always intrigued by the hardware side of computers and wanted to leave my future open to that too. Computer Engineering lands at a perfect balance between software related courses as well as some Electrical Engineering and Embedded Systems related courses. One of the first courses I had to take was CprE 185, taught by Dr. Thomas Daniels, which teaches the fundamentals of the C programming language, while incorporating it into a familiar application: DualShock 4 controllers. This required course also focused on independent guided learning in the labs, which I think is a great experience for first year engineering students.

Now, as I experience my final semester at Iowa State, I can look back at all of the courses and experiences that shaped me as the engineer and student I am today. Some classes were more enjoyable, such as CprE 310, CprE 281, and Math 267 where a lot of things clicked quickly and I was able to enjoy the curriculum. However, there were also courses that were much more challenging, namely CprE 381, where every lecture, lab, and exam demanded more from me than I expected, teaching me that struggle and problem solving is an essential part of being an engineer. Perhaps more importantly, it taught me to lean on others around me. Collaboration was a crucial aspect of that course, and working through difficult lab projects alongside my peers not only helped me survive the course, but taught me collaboration skills that other classes didn't.

The culmination of all my courses and experiences is what brought me to my final semester, working through a senior design capstone project in a team, where every team member has responsibilities and teamwork is crucial. My team is developing an upgraded quadcopter system, integrating a Raspberry Pi and camera module to equip it with real time computer vision capabilities. On the software side, my responsibilities are focused around bridging the gap between the software on the Raspberry Pi and the firmware on the base quadcopter. The hardware side of our project is upgrading the quadcopter with better motors, a custom PCB, and opening up a channel for the Pi to communicate with the firmware. A project like this is a prime example of how neither software nor hardware can work independently in an embedded system. Every decision made on the hardware side has a direct impact on the software side, and vice versa. For example, the upgraded motors and propellers require the quadcopter's internal PID controller to be tuned again, and the communication method we decide to have the Pi use to communicate with the drone's firmware must be accommodated with hardware improvements. In many ways, this capstone project has been the most realistic preview of how professional

engineering works; there are several components of a project that must work in parallel, which requires active communication and collaboration.

As I conclude my undergraduate journey at Iowa State, I begin to reflect on the journey that has brought me here. If I could go back, there are a few things I would approach differently. First, I would push myself to get involved in student clubs and organizations. Looking back, the connections and experiences that come with being involved in an academic organization are invaluable. Also, I would make a much stronger effort to build relationships with my professors and industry professionals. The resources and knowledge that faculty members carry extend far beyond what is covered during lecture, and I should have acted on those opportunities early on to build a network and connections. Networking with industry professionals through career fairs is also something I wish I had prioritized earlier, as it is never too early to get out there and make those relationships.

As I look toward my future as an engineer, I am aware that the learning does not stop with my degree. Every opportunity that comes my way is a chance to learn from mistakes and success. While I am still exploring exactly where my career path will take me, one of my primary goals as I enter the professional world is to develop as a leader. Throughout my time at Iowa State, I have had the opportunity to be an active teammate and collaborator, but I did not place myself as much as I should have into positions of leadership. Iowa State has given me a strong foundation of technical knowledge, resilience, and collaborative experience, and I am eager to build on top of it.